CRM On-Premise Server — Deployment Notes

Prepared 2026-10-01 · Sizing for about 6 users · Based on the project documentation and measurements taken on the development machine

1. Short answer

> **Yes — an ordinary office computer can be the server.** Six users (two of them read-only / monitoring) is a very small load. The whole system runs on one machine with Docker Compose.

The hardware figures below are my sizing, not figures from the documentation: `OD-03` (server specifications) is still open with the server administrator, and `OD-05` (expected daily workload) is undefined.

2. What was measured (development machine, idle)

Item	Measured
Runtime containers: PHP, nginx, PostgreSQL, Redis, 4 queue workers, scheduler	**~470 MB RAM**, ~0–1% CPU
Database size	**14 MB**
Stored files (8 real PDFs, largest ~340 KB)	**8.5 MB**
`crm-spa-build` (`vite build --watch`)	~1 GB — development only, not part of production
Meilisearch	Not running (compose profile `search`; Module 15 not built)

3. Recommended machine

Resource	Minimum	Recommended
CPU	4 cores	4–6 cores
RAM	8 GB	**16 GB**
Disk	128 GB SSD	**256–512 GB NVMe SSD**
Backup disk	Separate 500 GB	Separate 1 TB, or another machine (`BK-08`)
Network	Wired Ethernet	Wired Ethernet, static LAN IP
Power	UPS	UPS + BIOS set to power on after power loss

Why 16 GB: the idle stack is tiny, but a headless-Chrome PDF render, Meilisearch (not yet running) and PostgreSQL caching all spike. These spikes are estimates, not measured. Disk space matters mostly for backups (7 daily + 4 weekly copies, `BK-03`) and logs (`laravel.log` reached 1.44 GB during QA — fix F-26).

4. Architecture

Keep what already exists: **one machine running Docker Compose** — the production-matched setup that `D-66` calls for.

1. **OS:** Ubuntu Server LTS is recommended — it boots by itself and never forces a reboot. Windows + Docker Desktop can run the same containers (Arabic fonts and Chrome are baked into the image, `docker/php/Dockerfile`) but it is not built for 24/7 server use — see section 8.
2. **Compose:** run the existing `docker-compose.yml` with `restart: unless-stopped` on every service. The boot order PostgreSQL → Redis → Meilisearch → app → workers → nginx is already modelled (`ST-02`).

3. **LAN access:** nginx serves HTTPS on the company network — the primary path for everyone.
4. **External access:** `cloudflared` (Cloudflare Tunnel + Access) for the five named external users (`D-59`). The tunnel is outbound only, so no inbound port is opened.
5. **Backups:** the system makes the daily backup; it must be stored on a separate disk or machine (`BK-08`), with a real restore test every month (`BK-06`).
6. **Alerts:** an external heartbeat (`OBS-07`) — a "server down" alert cannot be sent from a server that is down.

5. Before deploying "as is"

- **Missing modules:** Modules 11–15 are untouched — no Procurement, Outdoor Visits, Reports, Dashboard or Meilisearch. There is also no WebSocket service in the compose file, although §14.2 lists one.
- **Pending fixes:** the post-QA fix pass F-17…F-30 and the limits test L.0–L.7 are still ahead.
- **Real-server test:** `P-02` has not run — a Hello page on the real server, reachable from a phone through Cloudflare. The deployment-debt register in `CHECKLIST.md` lists what it would prove.
- **Dev-only services:** compose contains `spa-build` (watch-mode build) and `boot-order-probe`. A production compose should build the SPA once and drop both.
- **Single point of failure:** `R-01` — one server, no standby. Backups and a tested restore are the only mitigation.

6. How to check an existing computer

Windows

Fastest: press `Ctrl+Shift+Esc` → **Performance** tab. CPU shows model and cores, Memory shows RAM, Disk shows capacity and **SSD or HDD**. Or in PowerShell:

```
Get-CimInstance Win32_Processor | Select-Object Name, NumberOfCores,
NumberOfLogicalProcessors
```

```
(Get-CimInstance Win32_PhysicalMemory | Measure-Object Capacity -Sum).Sum / 1GB
```

```
Get-PhysicalDisk | Select-Object FriendlyName, MediaType, Size, HealthStatus
```

macOS

```
system_profiler SPHardwareDataType SPStorageDataType
```

Linux

```
lscpu | grep -E "Model name|^CPU\(s\)"
```

```
free -h
```

```
lsblk -d -o NAME,SIZE,ROTA,MODEL      # ROTA 0 = SSD, 1 = spinning disk
```

Check	Good	Red flag
Disk type	**SSD or NVMe**	**HDD — PostgreSQL on a spinning disk is slow and fails more often**
RAM	**16 GB (8 GB is the floor)**	**4 GB or less**
CPU	**4+ cores, 64-bit, last ~8–10 years**	**2 cores, or very old**
Disk health	**Healthy**	**Warning / Caution (CrystalDiskInfo, or `smartctl -a`)**
Network	**Ethernet port**	**Wi-Fi only**
Free space	**128 GB or more**	**Almost full**

Cheap fixes: add RAM, or replace an HDD with a 256 GB SSD. A laptop also works (its battery acts as a small UPS) if sleep-on-lid-close is disabled and it is kept cool. In the BIOS set **Restore on AC Power Loss → Power On** (`ST-01`).

7. Existing data and virtualization

Short answer: you do **not** have to delete the old data if you run the system *on top of* the existing operating system. You **do** lose it if you install Linux directly onto the disk — which is the recommended route.

Route	Old data	What to know
A. Fresh Linux install **Recommended**	**Erased** on that disk	Cleanest and most reliable: starts at boot, no forced Windows updates. **Safer variant:** fit a new 256 GB SSD and keep the old disk untouched on a shelf — nothing to copy and nothing lost.
B. Virtualization or Docker Desktop on top of the existing OS (VM, or WSL2)	**Kept**	The VM or containers live in a virtual-disk file inside the free space. Costs: hardware virtualization (VT-x / AMD-V) must be on in the BIOS; the host OS keeps updating and rebooting; **Docker Desktop on Windows is not supported without a signed-in user, so it does not start by itself after a reboot** (conflicts with `ST-01`); about 1–2 GB extra RAM (estimate); old files and CRM data share one machine. Fine for a trial or a `P-02` test — not for 24/7 production.
C. Dual boot (Windows + Linux side by side)	**Kept**	Needs the partition shrunk, which can fail; after a power cut the machine may boot into the wrong OS. Not recommended for a server.

> **Rule for every route:** copy anything that matters to an external drive *before* touching the disk — even if you do not plan to erase it.

Docker on Linux (what this project uses) is container isolation, not hardware virtualization. Installing it adds only its own data folder (`/var/lib/docker`) and does not touch other files.

If you meant the CRM's own development data: production starts from an empty database — migrations and seeders create the schema, roles, permissions and limits. The development database (14 MB, test data) should not be copied blindly; if specific data must move, restore a dump deliberately with `pg_dump` / `pg_restore`.

Sources: project docs `CRM_Documentation_EN.md` §14.2, §16, `D-54`, `D-59`, `D-66`, `OD-03`, `OD-05`, `R-01`; `docker-compose.yml`; `docker/php/Dockerfile`; Docker community forum and Docker Engine docs on starting containers automatically.